

Thiago E. Kalid

MASTER'S STUDENT AT LASSIP

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Education

UTFPR (Federal University of Technology – Paraná)

M.SC. IN COMPUTER SCIENCE AND ELECTRICAL ENGINEERING

Curitiba, Brazil

Mar. 2025 - Present

- Advisor: Thiago A. R. Passarin
- Research topics: Ultrasonic imaging, Beamforming, Acoustic lens, Non-destructive Testing (NDT)

UP (University of Porto)

UNDERGRADUATE EXCHANGE STUDENT

Porto, Portugal

Aug. 2023 - Feb. 2024

- Completed classes: Audio Signal Processing, Machine Learning, Forecasting.

UTFPR (Federal University of Technology – Paraná)

B.SC. IN ELECTRICAL ENGINEERING (*Summa Cum Laude*)

Curitiba, Brazil

Mar. 2019 - Mar. 2025

- Thesis title: Development of a Visual Odometry Method for Ultrasonic Immersion Inspection
- Graduated 1st in class.

Research Experience

LASSIP - Laboratory of Statistical Signal Processing & Inverse Problems

GRADUATE RESEARCH ASSISTANT

Curitiba, Brazil

Mar. 2025 - Present

- **Advisor:** *Thiago A. R. Passarin*
- Developed a Python ultrasound simulator, based on ray-tracing, to support and optimize acoustic lens design.
- Designed and implemented a machine learning-based anomaly detection framework for flaw detection in ultrasonic non-destructive testing of pipes.
- Designed an acoustic lens capable of statically inspecting large pipe sections (e.g., 90°), significantly reducing inspection time in subsea inspections. The technology was applied and is currently being patented by the Brazilian Petroleum Company (Petrobras).

LASSIP - Laboratory of Statistical Signal Processing & Inverse Problems

UNDERGRADUATE RESEARCHER

Curitiba, Brazil

Apr. 2021 - Aug. 2023

- **Advisor:** *Thiago A. R. Passarin*
- Developed an image-processing-based system for underwater 2-D position estimation using a Raspberry Pi.
- Formulated and solved an optimization problem to compensate for ultrasound image distortions caused by temperature gradients in weld bead inspections.
- Implemented a parametric model to estimate acoustic refraction profiles between heterogeneous media using ultrasonic data.
- Implemented and evaluated multiple ultrasonic image reconstruction algorithms for non-destructive testing applications.

Laboratory of Electronic Control of Electrical Machines

UNDERGRADUATE RESEARCHER

Curitiba, Brazil

Apr. 2019 - Mar. 2021

- **Advisor:** *Walter D. Sanchez*
- Implemented numerical methods in MATLAB for electrical engineering applications, including power flow analysis and electrical machine modeling.
- Programmed, deployed, and tested electrical machine controllers (e.g., soft starters and PLCs) for industrial applications using WEG equipment.

Research Output

Peer-review publications

- [1] Pires, G. P[†], **Kalid, T. E.[†]**, de A. Prado, T., Costa, V. L., Pereira, G. R., Passarin, T. A. R., & Pipa, D. R. (2025). *An acoustic lens for displacement-free sectorial inspection of pipes with ultrasonic phased arrays*. *NDT & E International*, 156, 103459. <https://doi.org/10.1016/j.ndteint.2025.103459>. ([†] denotes equal contribution).
- [2] **Kalid, T. E.**, Everton Trento Jr, Tatiana A. Prado, Gustavo P. Pires, Giovanni A. Guarneri, Thiago A. R. Passarin, & Daniel R. Pipa. (2023). *Virtual encoder: a two-dimension visual odometer for NDT*. *Research and Review Journal of Nondestructive Testing*, 1(1). <https://doi.org/10.58286/28119>

[3] Muller, M. Y., de Almeida Prado, T., **Kalid, T. E.**, Passarin, T. A. R., & Pipa, D. R. (2023). *Ultrasonic sectorial inspection in the presence of temperature gradients*. *Research and Review Journal of Nondestructive Testing*, 1(1). <https://doi.org/10.58286/28122>

Publications in preparation

[1] **Kalid, T. E.**, Lazzaretti, A. E., de A. Prado, T., Pires, G. P., Pipa, D. R., & Passarin, T. A. R. (2026). *Machine learning-driven flaw detection for ultrasonic pipe inspections with acoustic lens*. Manuscript accepted for presentation at the European Conference on Non-Destructive Testing (ECNDT 2026), Verona, Italy.

[2] Wang, F. Z. Y., de A. Prado, T., **Kalid, T. E.**, Pires, G. P., Brante, G., Pipa, D. R., & Passarin, T. A. R. (2026). *Modeling and removal of electronic crosstalk effects in ultrasound signals by deconvolution*. Manuscript accepted for presentation at the European Conference on Non-Destructive Testing (ECNDT 2026), Verona, Italy.

[3] Costa, V. L., **Kalid, T. E.**, Dário, M. F., de A. Prado, T., Pires, G. P., Pipa, D. R., & Passarin, T. A. R. (2026). *Suppression of geometrical artifacts from acoustic lenses through low-rank approximation*. Manuscript accepted for presentation at the European Conference on Non-Destructive Testing (ECNDT 2026), Verona, Italy.

[4] Guarneri, G. A., Pegorini, V., Pipa, D. R., **Kalid, T. E.**, de A. Prado, T., Pires, G. P., & Passarin, T. A. R. (2026). *mini-auspex: a Python library for ultrasonic data import, post-processing and imaging*. Manuscript accepted for presentation at the European Conference on Non-Destructive Testing (ECNDT 2026), Verona, Italy.

[5] Dário, M. F., Caetano, F. H. R., Santin, D., Guerra, H. M., dos Santos, H. A., **Kalid, T. E.**, de A. Prado, T., Pires, G. P., & Passarin, T. A. R. (2026). *Robust virtual encoder: Using ultrasound data for compensation of depth variations in camera-based odometry*. Manuscript accepted for presentation at the European Conference on Non-Destructive Testing (ECNDT 2026), Verona, Italy.

Fellowships & Prizes

2025	3rd Best Thesis , Awarded by the Regional Council of Engineering (CREA-PR) for prominent electrical engineering thesis.	Paraná, Brazil
2025	Best in Class Honor , Awarded to the highest-achieving student (1st/60); ranked 8th among more than 1,600 Electrical Engineering graduates in the program’s history (Top 0.5%).	Paraná, Brazil
2025	Petrobras Graduate R&D Fellowship , Awarded by the Brazilian Petroleum Company (Petrobras) to support graduate research assistants at the R&D project AUSPEX.	Paraná, Brazil
2024	Ranked 1st , Selected as the top-ranked candidate for a full-tuition scholarship supporting an academic exchange period.	Paraná, Brazil
2023	PRH-ANP Undergraduate Fellowship , Awarded by the Brazilian National Agency of Petroleum, Natural Gas and Biofuels (ANP) to students with high academic performance and research potential in the energy sector	Brazil
2021	Petrobras Undergraduate R&D Fellowship , Awarded by the Brazilian Petroleum Company (Petrobras) to support undergraduate researchers at the R&D project AUSPEX	Paraná, Brazil

Skills

Programming & Computing	Python, MATLAB, C/C++, CUDA, NumPy, SciPy, Matplotlib, PyTorch
Tools & Platforms	Git, GitHub, Linux, $\LaTeX$, AutoCAD, SolidWorks
Additional Skills	Academic writing, Public speaking, 3D modelling and printing
Languages	Portuguese (Native), English (professional proficiency/C1), Spanish (B2)